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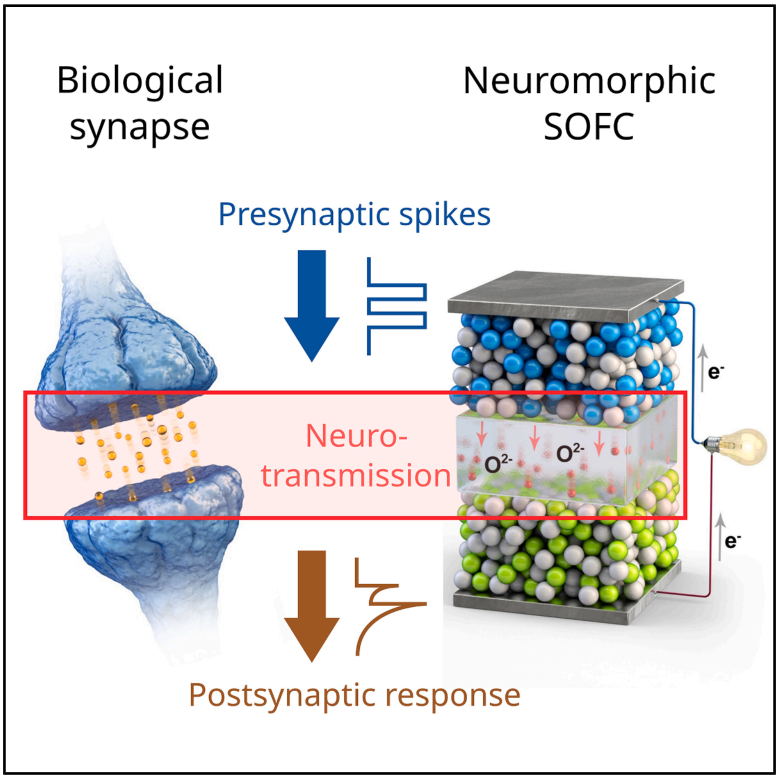
Article

In-cell neuromorphic computing in solid oxide fuel cells for bifunctional electrochemical power generation and artificial intelligence

Graphical abstract



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In brief

Seo et al. report that solid oxide fuel cells (SOFCs) exhibit synaptic-like behavior, enabling in-cell computing where computation and memory are integrated into the power generation process. This dual functionality supports applications in intelligent power management in smart grids, mobility systems, and adaptive and resilient energy infrastructures for sustainability.

Highlights

- We examine the potential of SOFCs in neuromorphic computing
- SOFCs exhibit synaptic properties, enabling brain-inspired computing systems
- Image classification via SOFC-based ANNs achieves a recognition accuracy of 85.4%
- SOFCs can manage power generation and cognitive computing, enhancing functionality

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